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Epidemiology of selected communicable diseases in Rwanda: a comprehensive analysis of the prevalence of major infectious diseases in Rwanda

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Abstract

Background Communicable diseases remain a major public health challenge in Rwanda, contributing significantly to morbidity and mortality. Despite notable progress in reducing the burden of infectious diseases such as malaria, HIV/AIDS, and tuberculosis, recent threats, including Marburg virus infections and other emerging outbreaks, underscore the need for renewed vigilance and strengthened disease control strategies. Transmission occurs through direct or indirect contact and is influenced by socioeconomic, environmental, and demographic factors. This study provides a comprehensive analysis of the prevalence of selected communicable diseases (Malaria, Tuberculosis, HIV/AIDS, Diarrheal Illnesses, and Neglected Tropical Diseases (NTDs) in Rwanda, highlights past successes, and proposes evidence-based recommendations for future control efforts.

Methods A comprehensive search was conducted using multiple sources, including PubMed/Medline, the World Health Organization (WHO) database, demographic surveys, scientific journals, the Centers for Disease Control and Prevention (CDC) databases, and Rwanda's Health Management Information System (HMIS). Studies were included if they were published between 2010 and 2024, focused on Africa, particularly Rwanda, and were available as free full-text. The most reported diseases across these studies were HIV/AIDS, malaria, tuberculosis, and diarrhoeal illnesses, reflecting their dominance in Rwanda's epidemiological profile. Findings relevant to the review objectives were summarised and organized thematically as a narrative literature review.

Results The incidence of malaria has dramatically decreased from 5.9 million in 2017 to 857,000 in 2022, highlighting the effectiveness of public health efforts in addressing this issue. However, HIV/AIDS continues to be a serious public health concern, with an adult prevalence rate of 3.0%. Although the distribution of highly active antiretroviral therapy (HAART) has been extremely successful, the programmes dependence on external funding threatens long-term sustainability. Inadequate sanitation and hygiene practices have resulted in high rates of diarrhoeal infections among children under five, most of which are caused by microorganisms in contaminated water.

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Conclusion Public health initiatives successfully implemented in Rwanda have controlled infectious diseases. The techniques used include providing long-lasting insecticide-treated nets, indoor residual spraying, engaging community health workers in detecting illnesses, and directly monitoring tuberculosis treatment adherence. However, to further reduce mortality, issues such as the need for sustained funding, the reliance of HIV/AIDS programmes on external financing, and antimicrobial resistance in malaria and tuberculosis must be addressed more effectively.

Clinical trial number Not applicable.

Keywords Epidemiology, Infectious diseases, HIV/AIDS, Rwanda, Healthcare

Introduction

Communicable diseases pose a significant health challenge, contributing to high morbidity and mortality rates and troubling population trends [1]. Harmful microorganisms cause these diseases, where transmission from living (direct) or non-living (indirect) sources to individuals can result in infection [1]. Among young people, infectious diseases are the leading cause of death [2], attributed to various modes of transmission, including water, air, human secretions, droplets, skin-to-skin contact, and food [2]. The interactions between humans and their environment's physical, social, and biological elements form the foundation for infectious disease transmission. Therefore, managing these factors can reduce the morbidity and mortality associated with communicable diseases worldwide [3].

The risk of transmission and burden of infectious diseases is particularly heightened in low- and middle-income countries (LMICs) and particularly in Sub-Saharan Africa due to confounding Environmental, Sociodemographic, Ecological, Infrastructural, and Behavioural risk factors [2, 3]. Among them, malaria, HIV/AIDS, tuberculosis (TB), and diarrhoeal diseases are especially burdensome in Sub-Saharan Africa (SSA), where health systems often face resource constraints, and social determinants amplify transmission risk [2]. In Rwanda, a country in east-central Africa, Infectious disease-related mortality is predominantly due to lower respiratory tract infections, HIV/AIDS, malaria, and diarrhoeal diseases, as shown in Table 1 [4]. While the country has made significant advancements in its health sector, it is still burdened by infectious diseases.

Therefore, studying the epidemiology of infectious diseases in Rwanda is essential. This study aims to provide a detailed analysis of Rwanda's current communicable disease profile with a focus on the prevalence of malaria, HIV/AIDS, tuberculosis (TB), and diarrhoeal diseases. It also aims to outline the socioeconomic and environmental determinants of infectious diseases in Rwanda, the surveillance and response systems to infectious diseases, as well as the challenges faced in the fight against infectious diseases by evaluating literature reviews, geographical and demographic data, randomised controlled trials, and other significant studies from various external sources.

Methodology

Search strategy

A comprehensive literature search was conducted using multiple sources, including PubMed/Medline, the World Health Organization (WHO) database, demographic surveys, scientific journals, the Centers for Disease Control and Prevention (CDC) databases, and Rwanda's Health Management Information System (HMIS) in July 2024. The most reported diseases across these studies were HIV/AIDS, malaria, tuberculosis, and diarrhoeal illnesses, reflecting their dominance in Rwanda's epidemiological profile.

To collate articles pertaining to epidemiology of infectious diseases in Rwanda a comprehensive search string using keywords: "infectious diseases Rwanda," "healthcare in Rwanda," "malaria," "HIV/AIDS Rwanda," "antiretroviral therapy," "tuberculosis Rwanda," "diarrhoeal diseases Rwanda," "neglected tropical diseases Rwanda," "environmental determinants," "disease control programs," "poverty and education," "urban and rural," "health care systems," "Rwanda," and "public healthcare" were devised. The Boolean operators "AND", "OR", and "NOT" were used to refine the search string.

Search criteria

Inclusion criteria

Studies were included if they were:

- Focused on HIV/AIDS, malaria, tuberculosis, and diarrhoeal illnesses.
- Were published between 2010 and 2024.
- Focused on Africa, particularly Rwanda.
- Were available as free full text.

Exclusion criteria

Studies were excluded if they were:

- Observational studies, news articles, comparative studies, conference proceedings.
- Not freely available as full-text papers.

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Table 1 List and distribution of top infectious diseases causing death in Rwanda [4]

Top Infectious Diseases Causing Death in Rwanda	Death Per 100,000 Individuals	References
Lower Respiratory Tract Infections	41.9	[4]
HIV/AIDS	22.4	[4]
Malaria	26.2	[4]
Diarrhoeal Diseases	24.8	[4]

Table 2 Summary of factors influencing the distribution of infectious diseases in Rwanda

Category	Examples of Contributing Factors	References
Environmental	Air and Water pollution, Climate change, changes in rainfall patterns, and vector habitats.	[6, 7]
Sociodemographic	Age, Gender, Education level, Income level, Urban vs. rural residence.	[8]
Ecological	Population size, Density, and Movement/migration patterns.	[9]
Infrastructural	Access to health care services, Sanitation facilities, and Clean water availability.	[8, 10]
Behavioural	Hygiene practices, Cultural beliefs, Delayed care-seeking, Self-treatment behaviours.	[8, 11]

- Not written in English.
- Not published between 2010 and 2024.

Data collection

Articles identified from PubMed/Medline, the World Health Organization (WHO) database, demographic surveys, scientific journals, the Centers for Disease Control and Prevention (CDC) databases, and Rwanda's Health Management Information System (HMIS) using the stated keywords were reviewed for title and the abstract in terms of their relevance to infectious disease in Rwanda applying the set inclusion and exclusion criteria by all authors. All collated articles with information relevant to the study and meeting the inclusion criteria were reviewed and analyzed. A narrative review was performed using the findings of the literature, and the findings were structured thematically. The selected articles were independently reviewed again by (L.D.B.) when reorganizing the findings and structuring the manuscript.

Results

Geographical and demographic factors influencing disease distribution

Rwanda is subject to a confluence of environmental (e.g., air and water pollution, climate change), sociodemographic (e.g., age, gender, education level, economic status, and place of residence), and ecological (e.g., population size, density, and composition) factors that impact the transmission of infectious diseases [5] (Table 2). These factors contribute directly and indirectly to the overall prevalence.

Environmental factors

Environmental conditions influence the transmission, survival, and distribution of infectious pathogens and vectors [6, 7]. This is particularly significant in mosquito breeding and malaria transmission as deforestation and climate change alter mosquito habitats and expand malaria risk areas. Contaminated water is also a major cause of diarrhoeal diseases by actively carrying pathogens.

Sociodemographic factors

Sociodemographic factors affect the exposure and health behaviours of the population, as low-income households often have crowded living conditions that increase the transmission of TB and diarrhoeal diseases [8]. These households also lack access to bed nets, clean water, sanitation, and healthcare. Young women are the demographic with the highest risk for HIV due to socio-cultural factors, while Migrant workers also have higher HIV, TB, and malaria exposure. Farmers and residents around Marshlands, wetlands, rice paddies, and stagnant water in general are at higher risk of exposure to Malaria.

Ecological factors

The interaction between humans, vectors, pathogens, and the environment in which they exist affects disease transmission through pathogen reservoirs, vector distribution, and ecosystem dynamics. Marshlands, wetlands, and rice paddies create high-risk areas for malaria by creating an ideal habitat for Anopheles mosquitoes [9]. A reduction in biodiversity due to factors such as deforestation and climate change can increase the population of vectors such as mosquitoes, as predation on mosquito larvae decreases, thereby increasing transmission of malaria [9].

Infrastructural factors

Healthcare and general infrastructure determine the prevention, detection, and response to infectious diseases. Poor WASH (Water, Sanitation, Hygiene) facilities drive diarrhoeal diseases [8]. Crowded living, working, and transportation conditions increase TB transmission. Limited diagnostic capacity leads to missed TB diagnoses. Limited road access reduces healthcare seeking and delays in diagnosis. Insufficient vector control and limited access to mosquito nets increase the risk of Malaria. promote malaria risk [10].

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Behavioural factors

Human behaviour influences exposure risk and healthcare-seeking behaviour. Handwashing reduces the risk of diarrhoeal disease [8]. Bed net use reduces malaria. Practicing unsafe sex, delayed testing, and poor ART adherence increase the transmission risk for HIV. Late presentation for TB testing and treatment, as well as lack of adherence to anti-TB medication, leads to disease progression, ongoing community transmission, and drug resistance. Poor ART adherence increases HIV viral load and transmission risk [11].

Major infectious diseases in Rwanda

Malaria

Prevalence in Rwanda

The western and southern provinces of Rwanda have shown a notable prevalence of malaria, with rates of 2.34% and 1.98%, respectively, indicating a high occurrence in these areas. In contrast, malaria prevalence rates in the eastern and northern provinces are lower, 1.35% and 1.45%, respectively [12]. The incidence of malaria in Rwanda decreased from 5.9 million in 2017 to 857,000 in 2022, a total reduction of 86% [13]. According to the President's Malaria Initiative 2017 data, 443,000 pregnant women and 1.8 million children under the age of five developed malaria in 2016. Malaria is endemic in 11 of Rwanda's 30 districts, with 19 districts prone to malaria epidemics [14].

Research indicates that individuals with lower incomes and education levels who lack insecticide-treated nets while sleeping, live in low-lying areas, do not have health insurance, and reside near irrigation zones face a higher malaria prevalence [12]. Various factors related to humans and mosquitoes influence the malaria infection rates. Higher temperatures and increased rainfall are expected to raise the prevalence of malaria. These weather conditions promote the breeding of various Anopheles species and facilitate the reproduction of parasites within mosquitoes. This phenomenon subsequently influences the risk of malaria transmission [15, 14]. Furthermore, urbanisation and agriculture may affect the spread of malaria. Urban areas typically have fewer breeding grounds for vectors; however, in certain situations, unsanitary urban conditions may encourage further reproduction [12, 16]. Malaria tends to be more prevalent in households with residents at lower elevations (i.e., below 1700 m above sea level). The southern districts of Rwanda have the lowest elevation levels, which may explain their higher malaria rates compared to other regions [13].

Public health interventions The distribution of long-lasting insecticidal nets (LLINs) or insecticide-treated mosquito nets (ITNs) may prevent malaria. Immunisation programmes, including mass campaigns, indoor residual spraying (IRS), intermittent preventive treatment of malaria in pregnancy (IPTp), seasonal malaria chemoprevention (SMC), and community-level use of rapid diagnostic tests (RDTs), have drastically reduced the incidence. Artemisinin-based combination therapies (ACTs) for *Plasmodium falciparum* have proved to be highly effective [17].

HIV/AIDS

Prevalence in Rwanda

In Rwanda, the adult HIV prevalence is 3.0% [18]. Estimates indicate that approximately 210,200 individuals in Rwanda are living with HIV, with a higher percentage of women (3.7%) compared to men (2.2%) [18, 19]. The prevalence of HIV among young adolescents is recorded at 0.4%, which suggests that about 5,900 young adolescents in Rwanda are estimated to be living with HIV out of a total of 216,000 people who are HIV positive [19]. HIV prevalence among adults in Rwanda varies significantly, ranging from 2.2% in the northern province to 4.3% in the city of Kigali. In urban areas, the prevalence of HIV is 1.9 times higher than in rural areas [19].

Public health interventions

The CDC aids in the scaling up of highly active antiretroviral therapy (HAART). Early testing and counselling at government-run institutions, distribution of self-test kits, voluntary medical male circumcision, and the prevention of mother-to-child vertical transmission have proven effective in Rwanda [18]. The CDC also provides pre-exposure prophylaxis (PrEP) for high-risk populations. These treatments and prevention efforts are essential for achieving HIV pandemic control by 2030 as a part of Rwanda's health agenda [20, 21]. An initiative that contributes to this goal is the DREAMS programme, which stands for Determined, Resilient, Empowered, AIDS-Free, Mentored, and Safe surveillance [21]. HIV incidence is at 0.08% among adults, because of the successful national HIV program [18].

Through a series of strategic decisions, Rwanda has successfully addressed HIV. Access to HAART has dramatically increased, and the nation has made significant progress in providing better care and support for all HIV-positive individuals. Fewer than 100 individuals were receiving HAART before 2002, while currently, over 150,000 individuals are undergoing treatment. The gradual increase of the CD4 threshold for treatment access contributed to this growth. In 2007, Rwanda achieved complete coverage of HAART with a CD4 cell count threshold of 200 cells/mm^3. The threshold was raised to ≤ 350 cells/mm^3 in 2008, and in 2013, it was further elevated to ≤ 500 cells/mm^3, with exceptions for urgent therapy for critical populations. Guidelines from 2015

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recommended that all patients be treated directly, regardless of their CD4 count. This expansion has led to significant advancements in the healthcare system, improving the overall health of the population [21].

Despite Rwanda's successful methods of reducing HIV incidence, progress has not been steady. External donors have provided 80% of the financial support for HIV prevention, treatment, and control. Still, such donations have been steadily declining, making it necessary to find new resources to finance the program. Strengthening human resources and optimising efficiencies will be crucial parts of Rwanda's strategy as it attempts to become more financially sustainable in providing HIV program services [21].

Tuberculosis (TB)

Prevalence in Rwanda

The first national TB prevalence survey in Rwanda was conducted in 2012, revealing an estimated 74.1 (95% CI 48.8–99.3) cases of smear-positive TB per 100,000 adults and 119 (95% CI 78.8-159.9) cases of bacteriologically confirmed TB per 100,000 adults [22]. By 2022, Rwanda's estimated TB incidence rate was 56 cases per 100,000 persons [23] (Clinical trial number not applicable).

Public health interventions

Poor outcomes of TB include treatment discontinuation for various reasons, such as developing adverse drug reactions and lack of patient follow-up [24]. Therefore, Directly Observed Therapy (DOT) is the most effective method to ensure treatment adherence. In DOT, a medical professional or another designated individual observes the patient infected with TB as they receive each dose of medication. Health professionals also enquire about any medication-related issues or adverse drug effects from patients during DOT treatment [25].

Rwanda has also developed a prevention strategy to foster harmony, ensure consistent management, and enhance program visibility. This strategy includes building the capacity of healthcare professionals and peer educators, conducting media campaigns, promoting health, and engaging with the community and stakeholders [24]. In recent years, Rwanda has succeeded in the fight against TB, where increased detection efforts led to a decline in TB cases from 7,230 in June 2011 to 5,763 in June 2016. Community health workers accounted for 46% of presumed TB cases referred to medical facilities and 25% of cases identified annually. Furthermore, 95% of those followed up by community health workers received adequate treatment [26].

TB screening in Rwanda has increased among high-risk populations. Three reference laboratories in Rwanda: the National Reference Laboratory, Kigali, and Butare University Teaching Hospitals have reliable diagnostic capabilities for multi-drug resistant-TB (MDR-TB). GeneXpert machines have improved TB detection and the early identification of MDR-TB cases. All national HIV care and treatment facilities now provide TB preventive treatment (TPT) services. Before completing TB therapy, 94% of co-infected TB/HIV patients received HAART. In 2020, Rwanda achieved a staggering 89% success rate in treating all forms of TB [24].

Diarrhoeal diseases

Prevalence in Rwanda

In developing countries, the two most common causes of diarrhoea are rotavirus and *Escherichia coli* [27]. The 2019 Rwanda Demographic and Health Survey (RDHS) reported a prevalence of diarrhoea of 14.3% among children under five [28]. This is a reduction from the 2014–2015 RDHS, whereby the prevalence was reported at 12.25% for the same age group [29]. Following the introduction of the Rotavirus vaccine in 2012, there was a reduction in rotavirus positivity of stool samples from ~ 35% in 2010 to ~ 19% by 2015, making E.coli the leading cause of diarrhoeal disease outside of outbreaks [30, 31]. While there have been endemic outbreaks post-2010, they are low-level and well-contained, with Vibrio cholerae being the leading causative pathogen [32]. The annual incidence rate of diarrhoeal illness in children under five years of age was 12,669 per 100,000 persons, with 1,012,827 cases reported between 2014 and 2018.

Public health interventions

A study on children under the age of 5 years revealed that the risk factors for developing diarrhoeal illness included a lack of clean water and inadequate hygiene and sanitation [33]. Additionally, children residing in areas with inappropriate amounts of faeces, those living on the street and dirty floors, and those not vaccinated against rotavirus were more susceptible to diarrhoeal illness. This finding was significant, as it influenced how healthcare initiatives would address childhood diarrhoeal illness in Rwanda. To improve household hygiene, the Ministry of Health (MOH) and other relevant parties should raise awareness about environmental sanitation, particularly the improper handling of human waste, such as faecal matter. Furthermore, mothers and other carers should be encouraged to vaccinate their children against rotavirus [34].

Previous research has demonstrated that children of highly educated mothers have a lower risk of diarrhoea-related morbidity and mortality due to their acquisition of knowledge regarding the maintenance of proper sanitation and hygiene [33, 35, 36]. As farmers use contaminated water for crop irrigation and food preparation, research found that families engaged in agricultural

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activity are more susceptible to diarrhoeal illness caused by soil-transmitted parasites [33].

The economic burden of diarrhoeal illness is greater for the country and the family [33]. A study conducted in Rwanda revealed that, despite 93% of the general population possessing health insurance, the annual cost of diarrhoeal illness was 101 US dollars per family [33].

Neglected tropical diseases (NTDs)

Prevalence in Rwanda

According to data from 2019, the most common neglected tropical diseases (NTDs) are intestinal worms, affecting 41% of the population with a 48% prevalence in adults; bilharzia schistosomiasis with 1,000 cases; podoconiosis with approximately 6,000 cases annually; scabies with 100,000 cases; taeniasis with 3,000 cases; cysticercosis; rabies from dog bites with 1,000 cases; and snakebite envenomation with 1,500 cases. Trachoma, lymphatic filariasis, mycetoma, and other less common pathologies are not as well-known as they once were, including onchocerciasis and human African trypanosomiasis *rhodesiense* [37, 38].

According to a 2014 nationwide follow-up mapping study on schistosomiasis (SCH) and soil-transmitted helminths (STH), the overall prevalence for any STH, including ascariasis, trichuriasis, and hookworm, was 45% [38]. Recent findings from mapping funded by the Centre for Global Health Research at Brighton & Sussex Medical School in the United Kingdom indicate podoconiosis is prevalent throughout Rwanda, with a national frequency of 68.5 per 100,000 individuals [39]. Almost every district in Rwanda reports cases, with the northern and western provinces being the most affected.

Scabies (*Sarcoptes scabiei var hominis)* is a mite that causes human scabies, a highly contagious dermatological condition. The most common symptoms include skin rashes and pruritus. In 2018, students from 8 primary schools participated in a cross-sectional survey, which found a prevalence of 1–20%. In Rwanda, the true prevalence and spread of scabies remain unknown. Regarding cysticercosis, a study conducted in a Rwandan health institution involving individuals with epilepsy in the southern district demonstrated that 23.3% of cases were related to neurocysticercosis in 2010 alone [40].

Following a Trachoma Rapid Assessment (TRA) study [38] conducted in nine districts from 2007 to 2008, the findings presented were as follows:

Trichiasis: Among the 20,417 individuals selected from 9 districts, the study identified 20 cases: 7 in Kayonza, 5 in Gicumbi, 3 in Nyagatare, 2 in Nyamasheke, 2 in Rulindo, and 1 in Nyaruguru. Corneal opacity developed in 5.2% of trichiasis cases, and one patient underwent surgical intervention. **Active Trachoma**: Out of 1,375 children screened, the study diagnosed 102 (7.4%). The entire district accounted for 5.8% of the cases, with one hamlet in Kirehe reporting 9 instances (18%), Gatsibo reporting 23 (15.3%) of the 150 children screened, and Nyaruguru recording 19 cases (12.6%) of the 151 children screened [38].

In 2008, two districts (Nyaruguru and Gatsibo) conducted the recommended prevalence survey to ensure a more representative sample. The findings showed that the prevalence of active trachoma infections in children aged 1 to 9 with follicular trachoma was 1.32% in Gatsibo and 0.73% in Nyaruguru. However, some villages had a prevalence higher than 5% (ranging from 0 to 13.2%), necessitating the WHO-recommended intervention of "mass drug administration" with azithromycin [38].

In 1999, a Rapid Epidemiological Mapping of Onchocerciasis (REMO) was conducted nationwide [38]. While the REMO found a few onchocerciasis nodules in 14 (14.3%) of the analysed villages, 84 (85.7%) showed no human onchocerciasis. In these villages, the distribution of the few cases of individuals carrying 1–3 nodules had highly sporadic distribution. They did not follow any identifiable patterns or river systems (i.e., potential breeding grounds or foci of transmission) that could have helped trace the source of the infections [38].

A lymphatic filariasis mapping survey was conducted in 2008 among adults over the age of 15 years from 13 selected villages across 5 districts (i.e., Bugesera, Rwamagana, Kayonza, Nyagatare, and Rusizi) identified as potentially endemic according to the WHO guidelines, where altitude is below 1500 m. The survey found only one positive case, 0.3% (1/296); a nocturnal blood smear confirmed the case (336). In 2017, a cross-sectional survey on the distribution and prevalence of podoconiosis in Rwanda [39–41], covering 80 sectors, administered filarial rapid diagnostic tests (Filariasis Test Strip-FTS and Wuchereria bancrofti-specific antigen-Wb123) to individuals with leg oedematous. Nine cases were positive across 6 districts (i.e., Rusizi, Nyamagabe, Gisagara, Gasabo, Ngoma, and Bugesera). Intestinal filariasis employs the Immunochromatographic Card Test (ICT) for diagnosis.

Envenomation by snake bite: In 2019, the HMIS in Rwanda reported over 1,000 snake bites. Dermatologists at referral hospitals found mycetoma rare in Rwanda, with fewer than five cases documented in the past five years [38].

Public health interventions

The MOH is working to treat and prevent NTDs by implementing various programmes. Many organisations, including the WHO and the CDC, collaborate with the MOH to conduct these initiatives aimed at preventing and managing:

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a. STH and SCH: Mass drug administration is conducted biannually for STH and annually for SCH. Health education for behaviour modification primarily occurs during deworming campaigns, utilising mass media communication and leaflets for on-site education. Community healthcare workers, totalling 42,000, are trained to support mass drug administration campaigns and education. Twenty SCH and STH surveillance sentinel sites monitor STH and SCH interventions, including drug efficacy. Collaboration between the public and private sectors is implemented to eradicate NTDs [38].
b. National treatment guidelines for trachoma, onchocerciasis, lymphatic filariasis, podoconiosis, scabies, and cysticercosis were established by integrating routine reporting in HMIS and disease mapping.
c. Rabies: Implementing the development of treatment guidelines, the incorporation of rabies among disorders under Integrated Disease Surveillance and Response (IDSR), and the availability of vaccination at the Rwanda Biomedical Centre (RBC), private pharmacies, and hospitals.
d. Envenomation by snakebite: development of medical guidelines, accessibility of anti-venom at the federal level and certain district pharmacies and raising public awareness of the pathology [38] (Table 3).

Surveillance and response systems

National surveillance systems for infectious diseases

Rwanda has established national surveillance systems to assist in monitoring and managing infectious diseases. The RBC, which operates under the MOH, oversees these systems by ensuring timely data collection, analysis, and dissemination [41, 42]. In 2002, Rwanda adopted a system denoted as the IDSR, which played a crucial role by integrating epidemiological surveillance and laboratory services to enhance early detection and response to disease outbreaks. Its primary focus is on prioritised communicable diseases, such as malaria, TB, HIV/AIDS, and emerging diseases like Ebola and COVID-19 [43].

Role of healthcare facilities and laboratories

In surveillance and response systems, healthcare facilities and laboratories play a pivotal role, whereby primary healthcare centres, hospitals, and specialised clinics report data to the national surveillance system. Laboratories, such as the National Reference Laboratory (NRL), provide diagnostic services and confirmatory testing, essential in ensuring comprehensive reporting of disease. Rwanda's health system is decentralised, allowing remote regions to contribute to the national data pool. The National Health Laboratory Network (NHLN) was established, which has reinforced and enhanced the country's capacity to monitor and respond to diseases [42].

Rapid response to disease outbreaks

In Rwanda, the preparedness plan for Ebola exemplifies the country's ability to respond quickly to disease outbreaks. Although no Ebola cases have been recorded, Rwanda has developed a comprehensive strategy that includes establishing isolation centres, training healthcare professionals, and conducting simulation exercises [44]. The Ebola virus disease preparedness plan covers border screening, community engagement, and the use of technology for real-time data sharing [45] (Table 4).

Socioeconomic and environmental determinants

In Rwanda, poverty and education levels notably affect the prevalence of diseases. Research shows that limited access to clean water, nutritious food, and adequate healthcare services due to financial constraints contributes to the spread of infectious diseases [51, 52]. Poor health literacy is also associated with lower education levels, reduced adherence to treatment protocols, and delayed diagnosis [52–54].

Table 3 Comparison between the infectious diseases prevalent in Africa

Diseases	Causative organism	Vector	Incubation Period	Vaccine Availability
Malaria	• *Plasmodium falciparum* (most common) • *Plasmodium malariae* • *Plasmodium vivax* • *Plasmodium ovale*	Anopheles mosquito	7–30 days (CDC)	• Malaria vaccines R21/Matrix-M • Malaria vaccine RTS, S/AS01 (RTS, S)
Tuberculosis	*Mycobacterium tuberculosis*	Air-borne disease (no vector)	Varies widely	The BCG (Bacillus Calmette-Guérin) vaccine
Acquired immunodeficiency syndrome (AIDS)	Human immunodeficiency virus (HIV)	Blood-borne disease (no vector)	2–4 weeks can be more (mayo)	Not available
Diarrheal diseases	• Rotavirus • *Escherichia Coli*	Water-borne disease (no vector)	2 days (Rotavirus) 3–8 days (E coli)	• Rotarix™ • RotaTeq™ • Rotavac™ and • RotaSiil™

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Table 4 National surveillance and response systems in African countries

Country	Surveillance System	Year of Implementation	References
Rwanda	Integrated Disease surveillance and response (IDSR)	2002	[46]
Uganda	Public Health Emergency Operations Center (PHEOC)	2013	[47]
Nigeria	Nigeria Centre for Disease Control (NCDC)	2011	[48]
South Africa	National Institute for Communicable Diseases (NICD)	2000	[49]
Kenya	Kenya Health Information System (KHIS)	2009	[50]

When comparing urban and rural populations in Rwanda, metropolitan regions, such as Kigali, benefit from greater access to healthcare resources because of their advanced medical facilities, improved infrastructure, and higher concentration of healthcare professionals. (50,451). In contrast, rural areas often encounter longer travel times, a shortage of healthcare practitioners, and limited transportation options, leading to higher disease prevalence and poorer health outcomes in these regions [55, 56].

The environment also influences the spread of infectious diseases through water quality, sanitation, and climate change. Research shows that in Rwanda, rainy seasons promote the reproduction of mosquitoes, resulting in increased malaria transmission rates. Also, poor sanitation and limited access to clean water contribute to the spreading of waterborne diseases such as cholera and dysentery. Furthermore, deforestation and changes in land use impact habitats of disease vectors, which also affect the prevalence of zoonoses [17, 57] (Fig. 1).

Challenges for disease control

Despite improvements in the healthcare system and government investments to establish a robust healthcare infrastructure, various challenges persist in Rwanda. Access to healthcare facilities is viewed as limited, especially in rural areas, reducing the chances for residents to receive care and benefit from medical services for disease prevention, detection, and early treatment [58]. The services provided by laboratories for early infection detection are scarce and underdeveloped in rural and district hospitals [59]. Furthermore, there is a significant disparity between healthcare providers in different regions of Rwanda. In some rural areas, a shortage of trained medical personnel, including nurses, doctors, and specialists, has resulted in extreme stress and a heavy burden on healthcare providers [60].

Despite having sufficient financial resources compared to other low-income countries, healthcare insurance in Rwanda does not fully cover all citizens [58]. Since the establishment of Rwanda's Community-Based Health Insurance (CBHI) programmes, approximately 90% of Rwandan citizens now possess health insurance; however, coverage relies on internal and external resources [58, 60]. This dependence on external donors to fund various disease control programmes undermines their progress

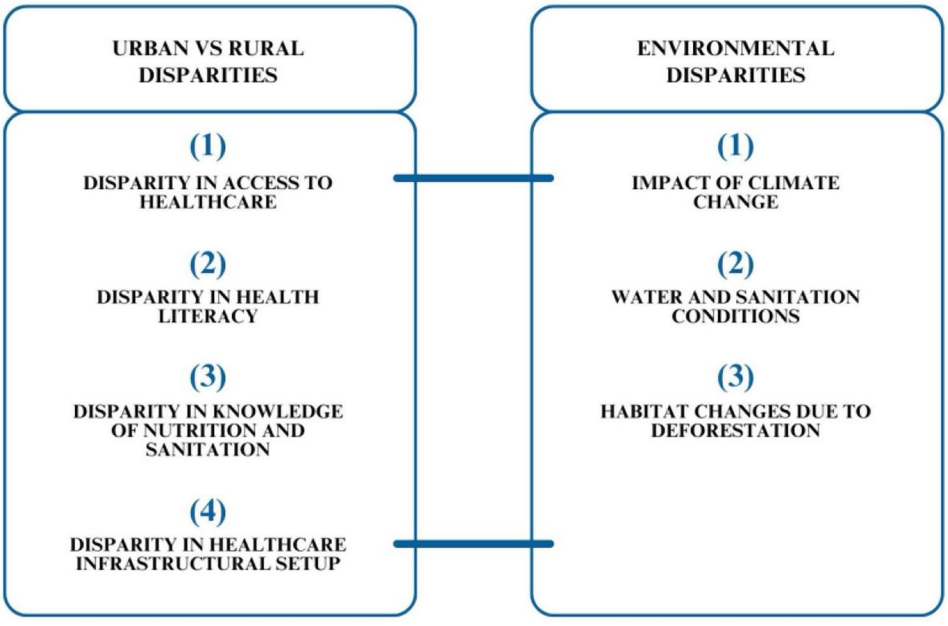

Fig. 1 Disparities in infectious disease control in Rwanda

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and success, risking their achievements [60]. Furthermore, resource limitations impact numerous laboratory findings, as the necessary equipment for conducting microbiological tests is inadequate. Consequently, there are delays in the detection of communicable diseases, leading to increased disease transmission and delays or failures in mitigation efforts [59, 60].

The political instability that Rwanda faced during the 1994 Genocide against the Tutsi still poses significant challenges to public health [61–63]. Cultural views, beliefs, and attitudes strengthen Rwandans' sense of self-worth and authority; however, this is undermined by their tendency to accept illness in its early stages and delay seeking treatment, particularly in men [63] (Fig. 2).

Success and innovations

Rwanda's achievements in addressing various communicable diseases have led to significant changes. Malaria, identified as one of the leading causes of death in the country and an important public health issue [64], has experienced a decline in both mortality and incidence rates, as reported by the WHO between 2016 and 2024 [65]. This reduction coincided with the ongoing successful advancements made by Rwanda's MOH through the National Malaria Control Program [66] and the extended National Malaria Strategic Plan (NMSP) [67]. With support from the CDC and the U.S. President's Malaria Initiative (PMI), the MOH has enhanced disease surveillance, detection, and treatment efforts. Additionally, the CDC provides antimalarial medications, pesticide-treated nets, and indoor residual insecticide spraying in high prevalence areas [20].

Rwanda has achieved impressive disease control for malaria and made significant strides in the fight against HIV. Over the past 25 years, the incidence and mortality rates of HIV/AIDS have decreased by 82% and 86%, respectively. In contrast, the life expectancy of HIV patients treated with HAART has increased by 25.6 years [68]. The collaboration among the U.S. President's Emergency Plan for AIDS Relief (PEPFAR), the CDC, and Rwanda's MOH has improved the quality of the services for preventing, detecting, and treating patients. Several measures have been implemented, including strengthening disease surveillance and increasing antiretroviral treatment [20]. The fight against TB in Rwanda boasts a nearly 90% success rate in treating all forms of the disease [69]. Moreover, the Rwandan government has embraced the 'END TB' strategy, which aims to reduce the financial and social impact of the disease by preventing and treating it to eradicate it by 2035 as a part of their agenda [70] (Table 5).

Rwanda has successfully addressed a range of infectious diseases and bio-threats by implementing various technologies. Launched in 1995, the Community Health Workers (CHWs) initiative has played a significant role in achieving the health-related Millennium Development Goals and enhancing primary health care in Rwanda, which has improved early infection detection and treatment [71]. Additionally, Rwanda adopted the One Health strategy in 2011. This approach took a multidisciplinary stance to tackle emerging challenges such as antibiotic resistance collectively, preventing and monitoring zoonotic disease outbreaks, and confronting other public health issues [72]. Nevertheless, the obstacles faced by

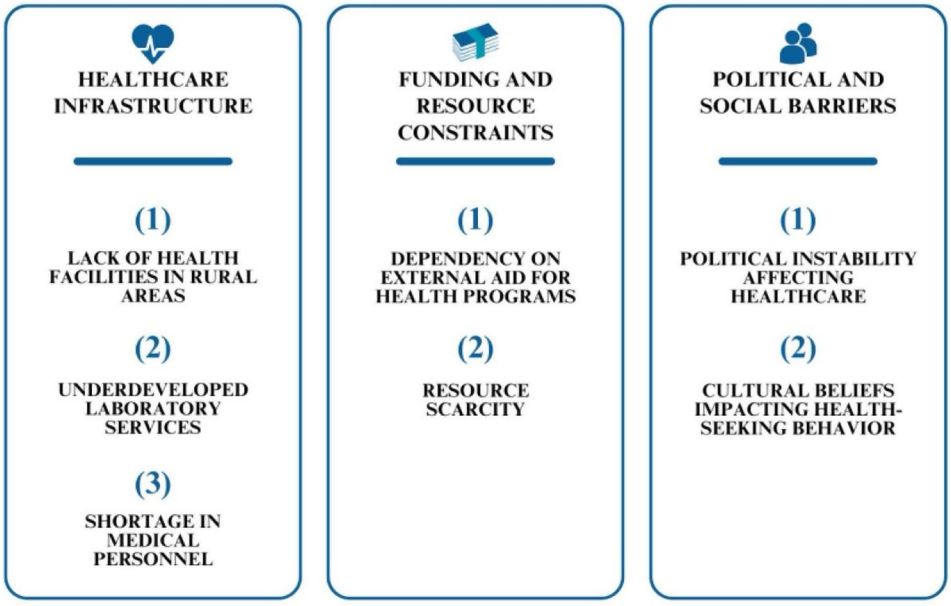

Fig. 2 Challenges for infectious disease control in Rwanda

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Table 5 Programs implemented to control infectious diseases in Rwanda

Disease	Program	Implemented by	Description
HIV/AIDS	HIV/AIDS Control Program	• Rwandan MOH • CDC • U.S. President's Emergency Plan for AIDS Relief (PEPFAR)	• Improves services offered to prevent, early detect, and treat patients. • Integrates different services for HIV testing and treatment at the MOH. • Enhances disease monitoring. • Intensifies antiretroviral therapy
Tuberculosis	END TB Strategy	World Health Assembly	• Decreases the financial and social impact of the disease • Prevents and treats TB to eliminate the global epidemic by 2035.
Malaria	National Malaria Program Extended National Malaria Strategic Plan	• Rwandan MOH • CDC • U.S. President's Malaria Initiative (PMI)	• Enhances disease surveillance, detection, and treatment. • Offers antimalarial drugs, pesticide-treated nets, and indoor residual spraying with insecticides for areas with high prevalence

Table 6 Innovative treatment and preventive strategies

Strategies	Description	Date Implemented
Community Health Workers (CHWs) Strategy	• Helps achieve the Millennium Development Goals for health. • Enhances Rwanda's primary health care, by facilitating early detection and treatment of infections.	Since 1995
Public-Private Partnerships (PPP) Strategy	• Launched to take advantage of investments from the private sector to improve health services and outcomes.	Since 2016
One Health Approach	• Prevents and monitors zoonotic diseases • Addresses emerging issues such as antimicrobial resistance • Targets other public health concerns collaboratively.	Since 2011

several disease control programmes stem from financial constraints and healthcare infrastructure challenges. These issues have prompted the Rwandan government to form public-private partnerships to harness private-sector investments to improve healthcare services and outcomes [73] (Table 6).

Future directions and recommendations

Given the current challenges, Rwanda must address them and decisively empower its healthcare system. Issues such as a growing population complicate the standardisation of a system that tackles this multifaceted problem, which affects various demographics and contributes to high morbidity and mortality rates [74–78]. The initiative to unify healthcare systems into a single-centred model called 'One Health' represents the latest effort to reduce Rwanda's infectious disease burden [77]. Launched in 2015, this initiative continues to streamline healthcare interventions and optimise resource utilisation by incorporating responses from the government, communities, and academia [78]. The 'One Health' policy aims to tackle factors contributing to poor health in Rwanda, including poverty, education, contaminated water sources, poor hygiene, gender inequality, and ongoing contact between humans and animals [79]. Future directions stemming from this policy include investing in clean water, sanitation, and hygiene (WaSH) to reduce the burden of diarrhoeal diseases and NTDs [80]. The WHO country cooperation strategic agenda reiterates in its 5th strategic priority that controlling pandemics and outbreaks requires building capacity for disaster risk management (DRM) within the health sector as well as enhancing the use of evidence for early warning, preparedness, and response to outbreak crises [81]. Addressing key challenges such as sustainable financing, antimicrobial resistance, and increasing preventive vaccinations and other preventive measures may keep Rwanda on track to become a leading example for the world in tackling infectious diseases.

In a tropical country like Rwanda, climate change and seasonality significantly impact the transmission patterns of infectious diseases [82]. Considering that the climatic conditions favourable for infections such as malaria and dengue fever are likely to become more common in rural areas compared to urban regions [83], these environmental effects could be minimised if financially viable interventions, along with other external factors, are implemented [84–86]. A 'One Health' approach also helps tackle this issue by harmonising the interactions among the environment, humans, and animals [80]. Additionally, preparedness for outbreaks and pandemics, achieved through the WHO strategic priority and enhanced sentinel surveillance, offers effective pathways for lessening the impact of climate change on infectious diseases [86, 87].

Ongoing investment in bolstering laboratory capabilities for diagnostic purposes and drug resistance monitoring, particularly for HIV, TB, and malaria, can be a game changer [88]. Additionally, exploring other socioeconomic and behavioural factors that influence the adoption of preventive measures, such as insecticide-treated nets and vaccinations, can help identify barriers and facilitators. Expanding access to essential medications, including antiretroviral therapy, anti-TB drugs, and insecticide-treated nets, is essential [89]. Moreover, increasing domestic funding for infectious disease

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control programs is necessary to lessen dependence on external donors, therefore ensuring financial stability [20] (Fig. 3).

Limitations

Readily available secondary data may contribute to inconsistencies in data collection methods and reporting periods and create conflicting evidence regarding similar variables. Additionally, we could only evaluate the overall burden of infectious diseases in Rwanda without being able to extrapolate or determine the long-term influence of specific interventions or programs. Data collection was also randomised, non-systematic, and based solely on available data.

Current status

Rwanda's epidemiological profile is characterised by high rates of lower respiratory tract infections, HIV/AIDS, malaria, and diarrhoeal diseases. The burden of malaria has significantly decreased, with reported cases dropping from 5.9 million in 2017 to 857,000 in 2022, demonstrating the effectiveness of public health interventions in tackling this problem. These interventions included the distribution of LLINs and indoor residual spraying. However, malaria remains endemic in several districts, particularly in the western and southern provinces, where socioeconomic factors, such as income and education, influence prevalence rates.

HIV/AIDS remains a major public health issue, with an adult prevalence rate of 3.0%. The expansion of HAART has been notably successful, but dependence on external funding poses risks to the sustainability of these programs. TB remains a concern, with an estimated incidence rate of 56 cases per 100,000 individuals in 2022, and drug-resistant strains complicate management efforts. Diarrhoeal diseases, mainly caused by pathogens in contaminated water, also exhibit high incidence rates among children under five years, worsened by insufficient sanitation and hygiene practices. NTDs, which include intestinal worms and SCH, further add to the disease burden, affecting a substantial portion of the population.

Public health interventions have proven effective in controlling infectious diseases. Strategies employed include the distribution of LLINs, indoor residual spraying, involvement of community health workers in disease detection, and directly observed therapy for adherence to TB treatment (Table 7).

Conclusion

Rwanda has achieved significant progress in alleviating the impact of communicable diseases through effective public health initiatives, collaborations across sectors, and active community involvement. Initiatives addressing malaria, HIV/AIDS, TB, diarrhoeal diseases, and neglected tropical diseases have produced significant outcomes, including an 86% decrease in malaria infections from 2017 to 2022 and an HIV/AIDS treatment success rate above 90% and marked improvement in diarrhoeal disease as a result of Rotavirus vaccination. These achievements highlight the efficacy of Rwanda's surveillance systems, community health worker programs, and preventive activities. Notwithstanding these advancements, Rwanda's disease control framework continues to be susceptible to systemic issues, including antimicrobial

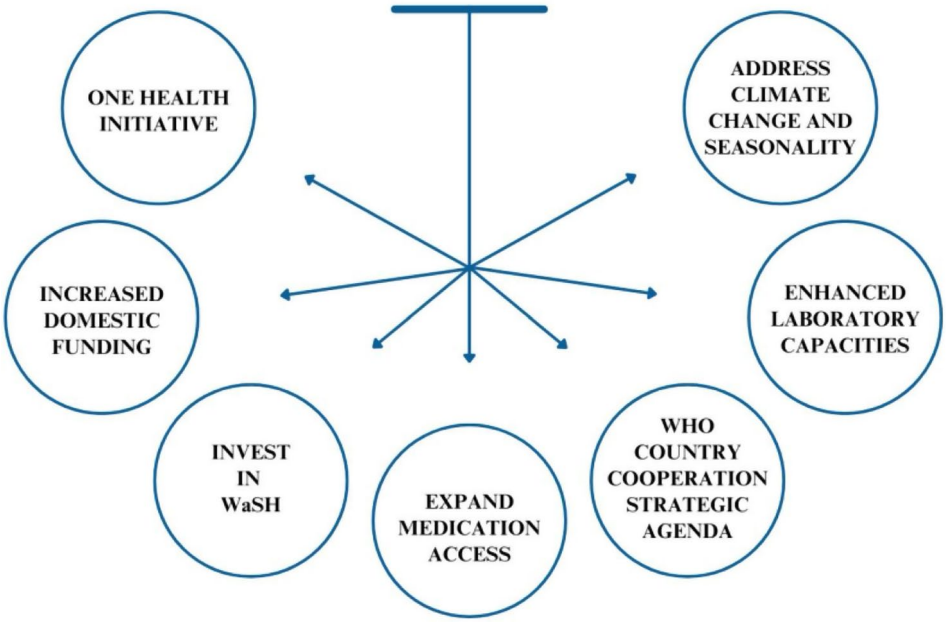

Fig. 3 Future directions and recommendations

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Table 7 Summary of key findings on infectious diseases in Rwanda

Infectious Diseases in Rwanda	Key Findings
Infectious Disease Burden/Causes	• Lower respiratory infections (41.9/100,000) • HIV/AIDS (22.4/100,000) • malaria (26.2/100,000) • Diarrheal diseases (24.8/100,000)
Malaria Trends	• 86% reduction in cases from 5.9 million (2017) to 857,000 (2022). • Remains endemic in 11 districts, with higher prevalence in Western and Southern Provinces (2.34% and 1.98%, respectively). • Influenced by socioeconomic factors (income and education levels). • Prevention strategies include: ° Long-lasting insecticidal nets (LLINs) ° Insecticide-treated mosquito nets (ITNs) ° Expanded program on Immunization ° Indoor residual spraying (IRS) ° Intermittent preventive treatment in pregnancy (IPTp), ° Seasonal malaria chemoprevention (SMC) ° Community-level use of rapid diagnostic tests (RDTs ° Treatment with artemisinin-based combinations (ACTs)
HIV/AIDS Trends	• Adult HIV prevalence is 3.0%, totalling up to 210,200 persons. • Higher rates in women (3.7%) than men (2.2%). • Prevalence in urban areas is 1.9 times greater than in rural areas. • CDC initiatives targeted ART scaling and prevention of mother-to-child transmission by campaigns like the DREAMS program. • Using ART and promoting adherence to it was achieved by a gradual increment in the CD4 threshold required for treatment access starting at 200 cells/mm^3 in 2007 reaching ≤ 500 cells/mm^3 in 2013.
Tuberculosis Trends	• Incidence rate of 56 cases per 100,000 people. • Smear-positive TB prevalence of 74.1 per 100,000 adults. • Bacteriologically confirmed TB prevalence of 119 per 100,000 adults. • Drug-resistant TB is becoming more prevalent. • Community health workers constituted 46% of TB cases referred to medical facilities and 25% of annually identified cases. • Antiretroviral therapy is being administered to 94% of co-infected TB/HIV patients before completing TB treatment regimens. • Frequent screening and early detection decreased TB cases from 7230 in June 2011 to 5763 in June 2016
Diarrheal Disease Trends	• Incidence rate in children under five was 12,669 per 100,000 between 2014 and 2018. • Annual cost of diarrhoeal sickness totals to 101 US dollars per family • Risk factors include lacking clean water and inadequate hygiene.
Neglected Tropical Diseases Trends	• Common NTDs include: ° intestinal worms (48% prevalence in adults) ° Schistosomiasis (1,000 cases annually) ° Podoconiosis (6,000 cases annually) ° Scabies (100,000 cases annually) ° Taeniasis (3,000 cases annually) ° Rabies from dog bites (1,000 cases annually) ° Snakebite envenomation (1,500 cases annually) • Soil-transmitted helminths (STH) prevalence was 45% in 2014. • Over 500 documented dog bite incidents since 2016. • Active Trachoma had a 7.4% prevalence in children (2008). • 20 cases of Trichiasis detected out of 20,417 people surveyed. • Mapping survey for Lymphatic Filariasis in 2008 found only one positive case (0.3%). • Filarial tests in 2017 found 9 positive cases in 6 districts. • Less than 5 cases of Mycetoma in the previous 5 years. • 14.3% of villages had a small number of Onchocerciasis cases in 1999
Public Health Maintenance strategies	• Integrated Disease Surveillance and Response (IDSR) started in 2002. • Ebola Preparedness Plan • Isolation centres • Training of healthcare professionals • Simulation exercises • Border screening

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resistance, reliance on external financing, climate fluctuations, and inadequate health infrastructure in rural areas. Policymakers must prioritise heightened local investment in health systems to diminish donor dependency, especially for HIV and TB programs, to maintain resilience and advancement. Enhancing universal health coverage and improving laboratory and diagnostic infrastructure are essential for prompt disease identification and response. Researchers are urged to examine context-specific behavioural and environmental determinants influencing disease transmission and the adoption of preventative measures such as insecticide-treated nets (ITNs), antiretroviral therapy (ART), and vaccines. Operational research must guide culturally relevant, evidence-based actions to enhance effect and sustainability. Public health professionals must enhance community involvement and health education initiatives, prioritising hygiene practices, vaccinations, prompt treatment-seeking behaviours, and integrated disease management within the One Health framework. To maintain progress and enhance Rwanda's disease control initiatives, it is imperative to institutionalise surveillance, foster local innovation, and fortify the health system's resilience against emerging infectious threats. Investing in good water, sanitation, health education, and climate-resilient policies will protect public health advancements. Rwanda's experience exemplifies a blueprint for other low-income nations, illustrating that political dedication, cohesive planning, and community-driven efforts may significantly enhance public health results. Through intentional efforts and ongoing cooperation, Rwanda is poised to emerge as a regional leader in the management and preparedness for communicable diseases.

Abbreviations

ACTs	Artemisinin-based Combination Therapies
AIDS	Acquired Immunodeficiency Syndrome
ANC	Ante-Natal Care
ART	Anti-retroviral Treatment
CBHI	Community-Based Health Insurance
CDC	Centre for Disease Control and Prevention
DHIS2	District Health Information Software 2
DOT	Directly Observed Therapy
DREAMS	Determined, Resilient, Empowered, AIDS-Free, Mentored, and Safe
EBV	Ebola virus
HIV	Human Immunodeficiency Virus
HMIS	Health Management Information System
ICT	Immuno-chromatographic Card Test
IDSR	Integrated Disease Surveillance and Response
IPTp	Intermittent Preventive Treatment of Malaria in Pregnancy
IRS	Indoor Residual Spraying
ITNs	Insecticide-Treated Mosquito Nets
KHIS	Kenya Health Information System
LLINs	Long-Lasting Insecticidal Nets
MDR-TB	Multi-Drug Resistance-TB
MOH	Ministry of Health
NCDC	Nigeria Centre for Disease Control
NHLN	National Health Laboratory Network
NICD	National Institute for Communicable Diseases
NMSP	National Malaria Strategic Plan
NRL	National Reference Laboratory
NTDs	Neglected Tropical Diseases
PEPFAR	President's Emergency Plan for AIDS Relief
PHEOC	Public Health Emergency Operations Centre
PLHIV	People living with HIV
PMI	President's Malaria Initiative
PPP	Public-Private Partnerships
RBC	Rwanda Biomedical Center
RDTs	Rapid Diagnostic Tests
REMO	Rapid Epidemiological Mapping of Onchocerciasis
SCH	Schistosomiasis
SMC	Seasonal Malaria Chemoprevention
SSA	Sub-Saharan Africa
STH	Soil-Transmitted Helminth
TB	Tuberculosis
TPT	TB Preventive Treatment
TRA	Trachoma Rapid Assessment
WaSH	Water, Sanitation, and Hygiene
WHO	World Health Organization

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